

B. Additional experiments for ICML 2026 rebuttals

B.1. Training, validation, and test performance during training

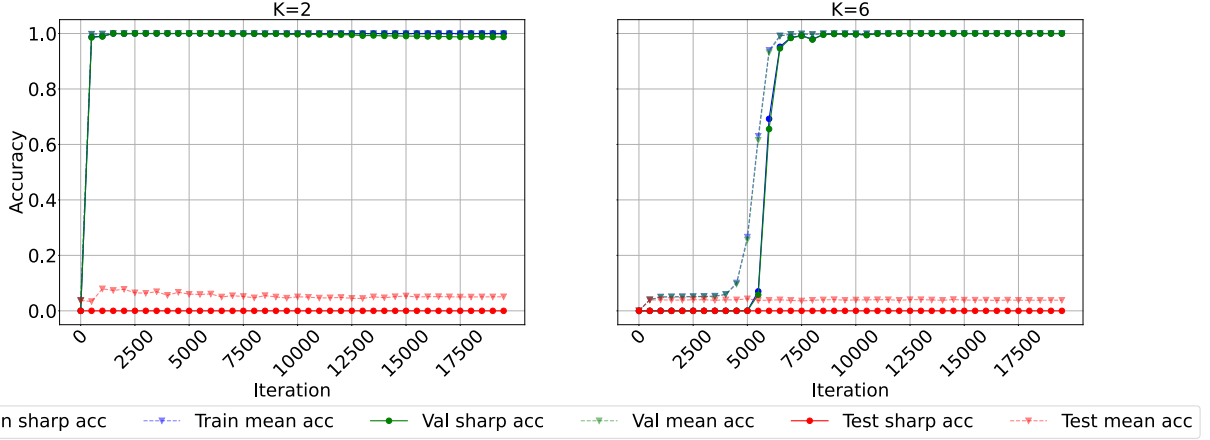


Figure 25. **Uniform benchmark (without identity), $N_heads = 6$, $N_layers = 3$, Direct (Relative)**: We observe that the models learn to predict the outputs of training compositions for both $K=2$ and for $K=6$, while the test accuracy on unseen compositions remains 0% throughout. We report two metrics: (1) **sharp accuracy**, where a prediction scores 1.0 only if the entire output sequence is correct and 0.0 otherwise; and (2) **mean token accuracy**, average accuracy of predictions in the output sequence, with both metrics averaged across compositions from each split.

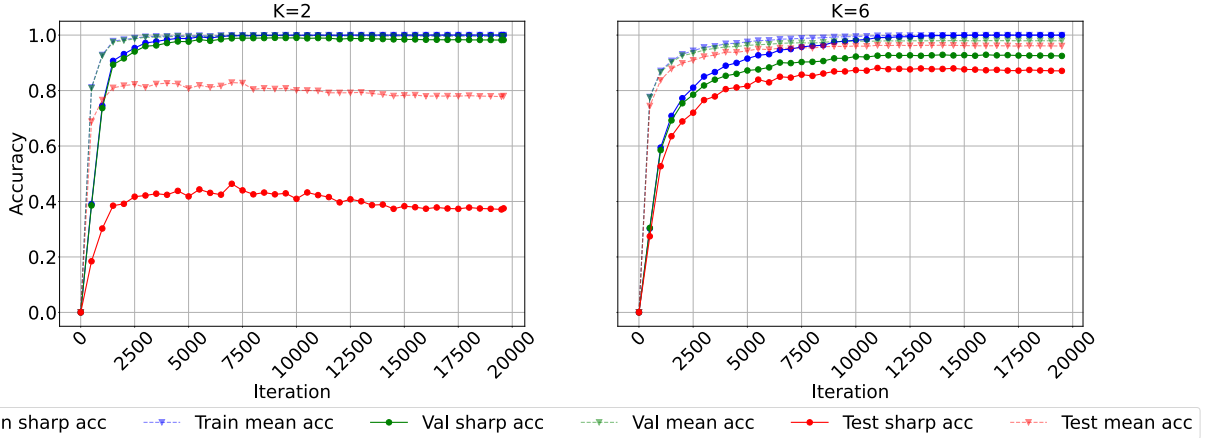


Figure 26. **Diverse benchmark, $N_heads = 6$, $N_layers = 3$, Direct (Relative)**: We observe non-trivial compositional generalization, as the models perform well on training and validation compositions, the test performance also increases. We observe better generalization for $K = 6$ than for $K = 2$ due to more training diversity and shared equivalences.